Logarithmic and exponential forms of equations

1

a $\log_2 32 = 5$ **b** $\log_6 36 = x$ **c** $\log_9 9 = 1$ **d** $\log_2 1 = 0$

e $\log_{0.9} 1 = 0$ **f** $\log_{16} 64 = 1.5$ **g** $\log_9 27 = \frac{3}{2}$ **h** $\log_{10} 0.01 = -2$

i $\log_3 \frac{1}{81} = -4$ **j** $\log_x 125 = 1.5$

2

a $4^2 = 16$ **b** $5^1 = 5$ **c** $8^0 = 1$ **d** $2^{-3} = 0.125$

e $(5.8)^2 = 33.64$ **f** $\left(\frac{1}{3}\right)^{-2} = 9$ **g** $x^5 = 32$ **h** $8^6 = x$

i $3^{-1} = \frac{1}{3}$ **j** $9^{\frac{1}{2}} = 3$

3 True

4 True

Inverse relationship between exponentials and logarithms

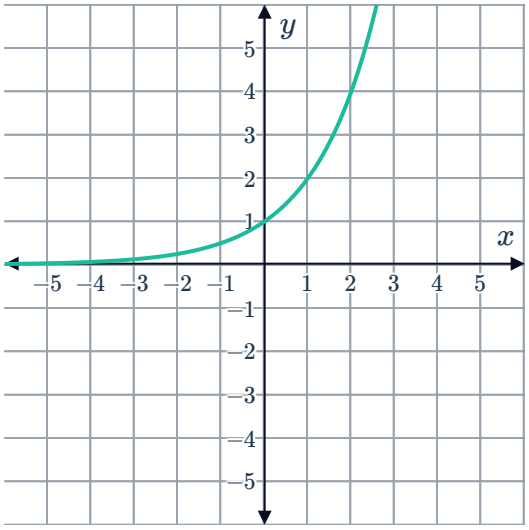
5

a Yes

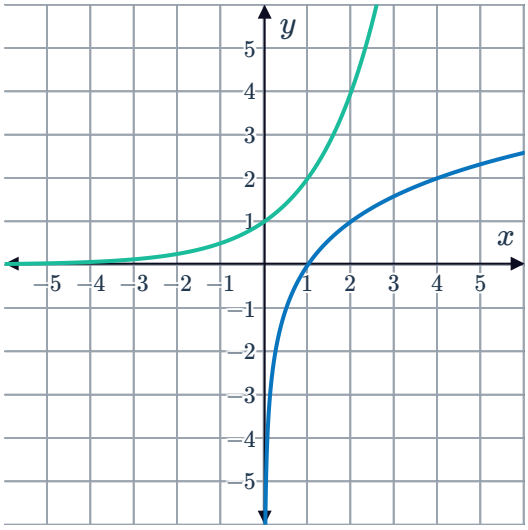
b $y = \log_{10} x$

6

a



b

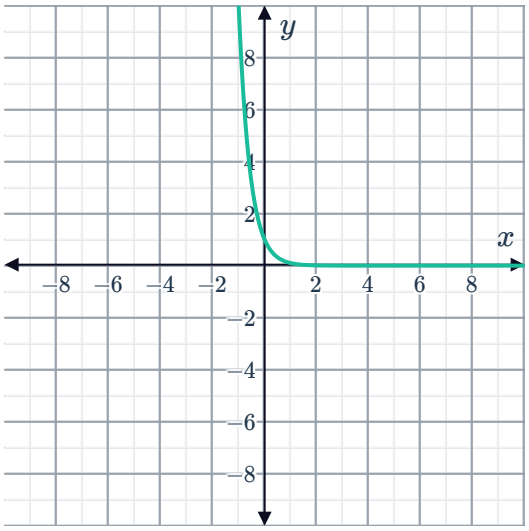


c x-intercept

© ClipSAT: By Mr. Mohamed Abdallah +966597688647

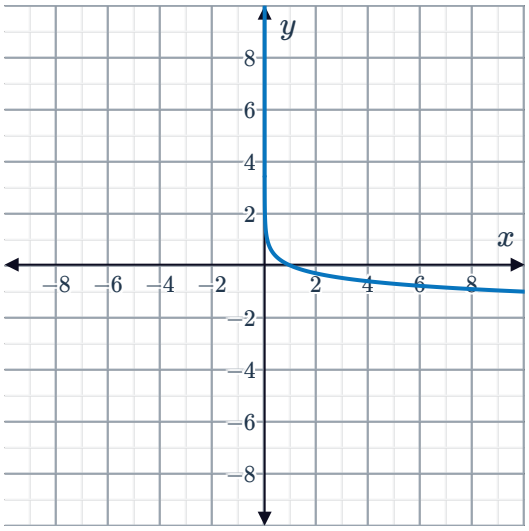
7

a



b $y = -\log x$

c



8

a $y = \log_e (x + 3)$ **b** $x > -3$ **c** $(-\infty, \infty)$

9

a $y = 5^x + 3$ **b** $(-\infty, \infty)$ **c** $(3, \infty)$

10

a $y = \log_e \left(\dfrac{x+5}{2} \right)$ **b** $(-4.986, -4.986)\,,\ (1.118, 1.118)$

© ClipSAT: By Mr. Mohamed Abdallah +966597688647